The UCSC Xena platform for public and private cancer genomics data visualization and interpretation

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Xena enables you to explore both public and your own multi-omic cancer genomics data

- 40 cancer types + pan-cancer
- **TCGA, Pan-Cancer Atlas,** GDC, PCAWG, ICGC, **GTEx normal vs TCGA tumor**, MET500
- Differential expression analysis, KM analysis, violin plots, subgroup by *any* data
- Easily and securely **view your own data** or data from the literature. Mac, Windows and Linux.

http://xena.ucsc.edu

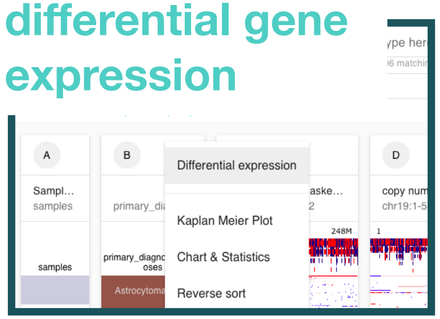
Highlights

Chart View

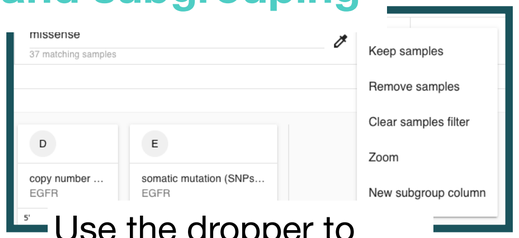
Violin plots

High quality PDFs

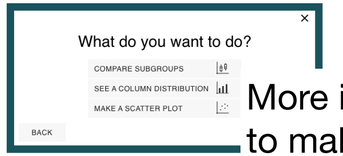
Genome-wide differential gene expression



Better filtering and subgrouping

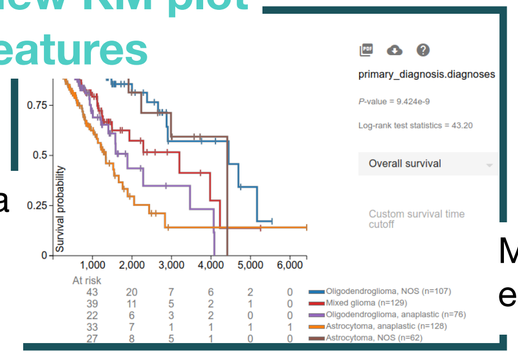


Use the dropper to select samples directly from the screen



More intuitive way to make charts

New KM plot features



Download data to reproduce

At risk table

Excel File support

Multiple survival endpoints

Xena can help you answer ...

- *Is the over-expression of gene A associated with lower survival in these two cancer types?*
- *Is gene B differentially expressed in tumor versus normal?*
- *What are the most differentially expressed genes for the subgroups I just made?*
- *What is the relationship between expression, mutation, copy number, etc for these genes?*

Data Types we support

- Somatic mutation (SNPs & INDELS)
- Expression: mRNA, miRNA, protein, lncRNA, gene, exon, transcript, etc
- Gene-level & segmented CNV
- Structural variants
- DNA methylation
- ATAC-seq peak signal
- Coding and non-coding regions
- Phenotype/Clinical (e.g. survival)
- Genomic signatures

Interested in a free workshop?

We provide online workshops free of charge to any institution or group! Email us at genome-cancer@soe.ucsc.edu for more information.

If you use us, cite us

Goldman, M.J., et al. Visualizing and interpreting cancer genomics data via the Xena platform. Nat Biotechnol (2020).